

Reflect your health issues before they really occur

The problem

Most adults with Type 2 diabetes in India live between clinic visits with a glucometer and a few finger-pricks, not a continuous glucose monitor.

The spike after a rice-heavy dinner happens when nobody is measuring, and the tools that could warn about it assume a daily sensor, Western food and English.

Team Pratifalan · Koushik Deb (solo) · Indian Institute of Information Technology Kalyani

Hosted demo: <https://pratifalan.duckdns.org>

Glucose twins are built for people who wear a sensor every day

Sensor

- Most twins re-read a CGM every few minutes.
- A continuous sensor is a recurring cost most patients here do not carry.
- Finger-pricks are what people already have.

Food

- Thali, katori, roti count: not grams on a label.
- No open Indian CGM dataset exists to learn from.
- A vision LLM's carb guesses barely tracked logs (Spearman 0.15, 78 photos).

Language

- Advice apps speak English.
- People ask in Bengali, Hindi, Kannada.
- They need a short spoken answer with numbers they can trust.

backend/reports/meal_photo.json (2026-10-05 17:23): US CGMacros meal photos; see slide 10 for the forecast cost.

A sensor-light glucose twin with Indian meal inputs, multilingual interaction and visible uncertainty

Target outcome: $P(\text{glucose} > 180 \text{ mg/dL within 2 h after a meal, for adults with T2D, with a 90\% band. } P(<70) \text{ is shown but labelled not validated.}$

Calibrate

an initial 5-day CGM wear
+ EHR: age, sex, BMI, HbA1c
(a clinic-lent sensor)



Keep it alive

about 2 finger-pricks a day
Indian meals -> food table carbs
activity; lab uploads



See, ask, learn

$P(>180 \text{ mg/dL in the next 2 h})$
two possible futures, 90% band
watch it learn from a reading
ask in bn / hi / kn / en

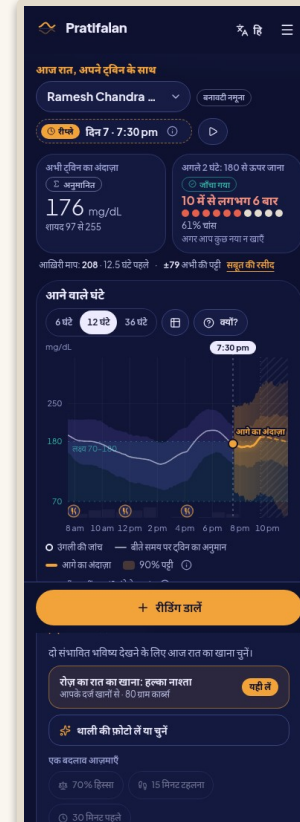
Tested setup: a 5-day CGM calibration wear, then finger-pricks. $P(<70)$ not validated: 83 low windows from 9 of 45 people (backend/reports/sensor_ladder.json, low_events).

Tonight, with your twin

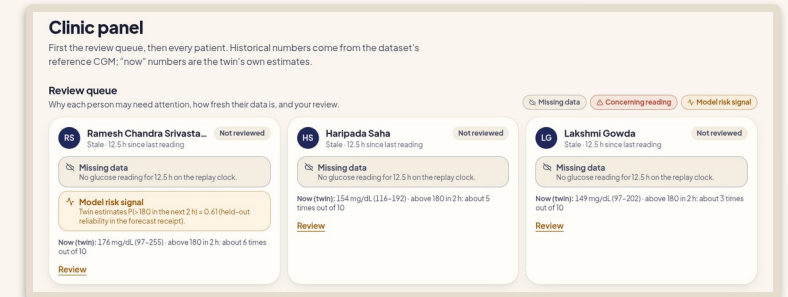
<https://pratifalan.duckdns.org> · personas are synthetic composites the production model never saw



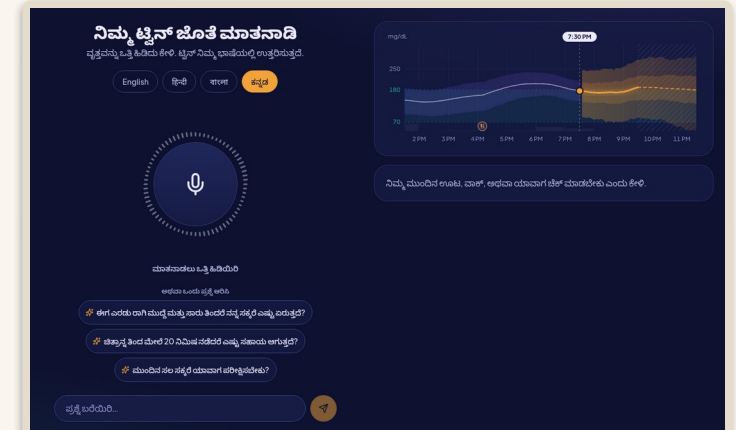
Twin stage: replay clock, risk with held-out reliability, Watch the twin learn



Hindi, 390 px



Clinician review queue



Talk in Kannada

Physiology first, learning second, honesty last

60-min RMSE, 2 pricks/day; paired difference vs the hybrid (95% CI)

Hybrid twin **30.6**

LightGBM, sparse **31.0**
−0.40 (−1.65 to 0.67)

Mechanistic only **33.7**
−3.11 (−4.19 to −1.87)

Personal average **33.3**
−2.69 (−4.35 to −1.10)

Persistence **38.9**
−8.31 (−10.83 to −5.02)

Level with LightGBM at 2 pricks/day (CI includes 0). Raw particle P(>180) ECE 0.125 → 0.036 after stacking + isotonic.

Mechanistic layer

7-state glucose-insulin ODE: own insulin secretion, 2-compartment gut, exercise uptake, dawn effect, drift. RK4 at 5 min over particles.

Personalisation

EHR-conditioned prior → CEM fit on the 5-day CGM wear → Liu-West particle filter; finger-pricks: Gaussian likelihood, adaptive inflation.

Learned correction

LightGBM residual (monotone in carbs) · normalised split conformal bands · stacked event classifier + isotonic · abstain.

Decisions

Paired what-if (Two possible futures) on the same particles · next best prick (EnSRF, experimental) · drivers: physiology vs learned.

mg/dL; backend/reports/baselines.json (paired_vs_full_hybrid_60min, 2026-10-05 17:07), calibration_curve.json. Same forecast windows for every method; nested person-wise CV.

2 pricks a day after 5 CGM days: 60-min error 30.6 mg/dL, band covers 89.9%

Full CGM 27.3 • no glucose data 31.1 • band width 114.4 mg/dL • AUROC P(>180) 0.81 (0.73–0.88)

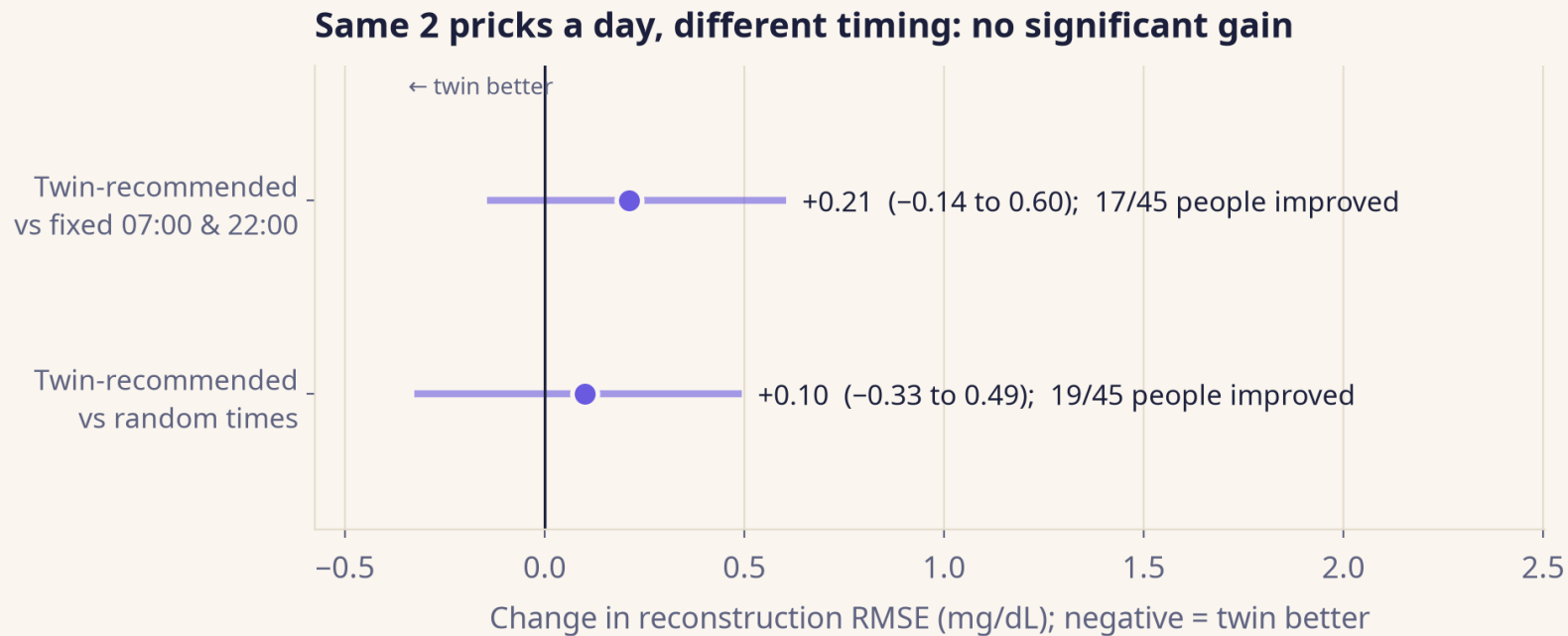


Source: backend/reports/sensor_ladder.json (2026-10-05 17:06). CGMacros, 45 adults, 5-day CGM calibration, nested person-wise 5-fold CV, 95% CIs by bootstrap over people; pricks simulated from the hidden CGM.

Living uncertainty, controlled: within 90 min of a prick the band is −10.6 mg/dL vs no data at 60 min (−17.5 to −4.7); 8 h or more later it is +12.0 (10.2 to 14.1). Recent prick: 60-min error −2.0 mg/dL (−3.8 to −0.5).

Choosing when to prick did not beat fixed clinic times

An honest null result, kept in the app only as an experimental suggestion



Source: backend/reports/nbp_value.json (2026-10-05 17:07). Paired over people, 95% bootstrap CI.

30.7

60-min RMSE, twin-recommended times

30.6

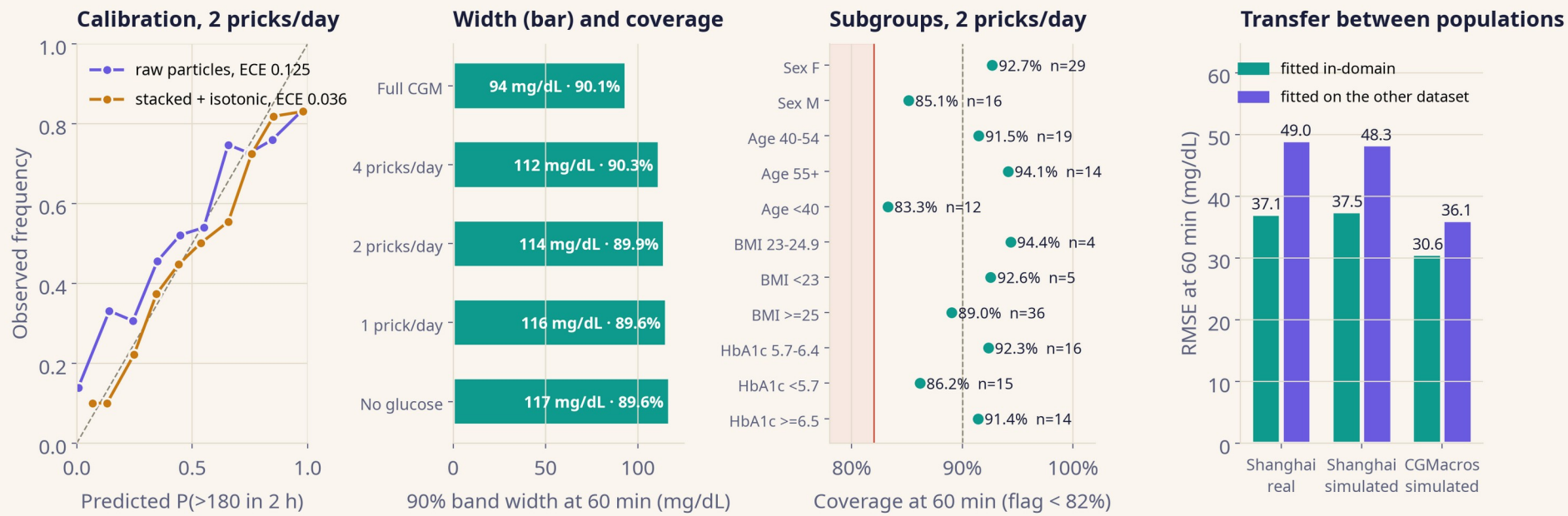
60-min RMSE, fixed 07:00 & 22:00

30.6

60-min RMSE, random times

backend/reports/nbp_value.json (2026-10-05 17:07); planner: ensemble square-root Kalman variance reduction over 12 h; 2 pricks/day in every arm.

Calibrated and covered; transfer to Shanghai costs 10.8 mg/dL



Sources: backend/reports/calibration_curve.json, sensor_ladder.json, subgroups.json, transfer.json (2026-10-05 17:14). Held-out predictions from the nested outer folds. ShanghaiT2DM: 100 people, 109 recordings; CIs resample people

89.9% · 114.4

coverage · width (mg/dL) of the 90% band at 60 min, 2 pricks/day

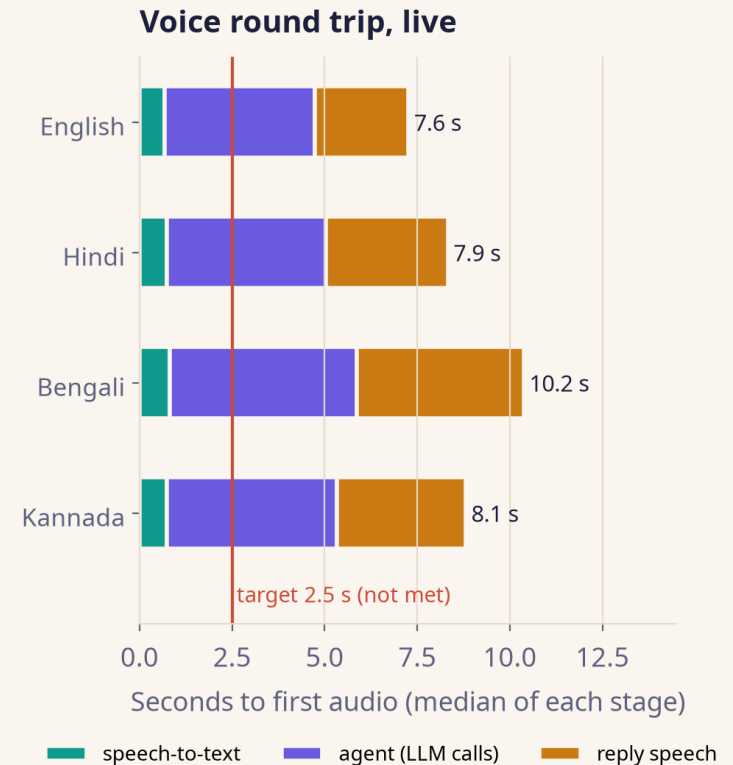
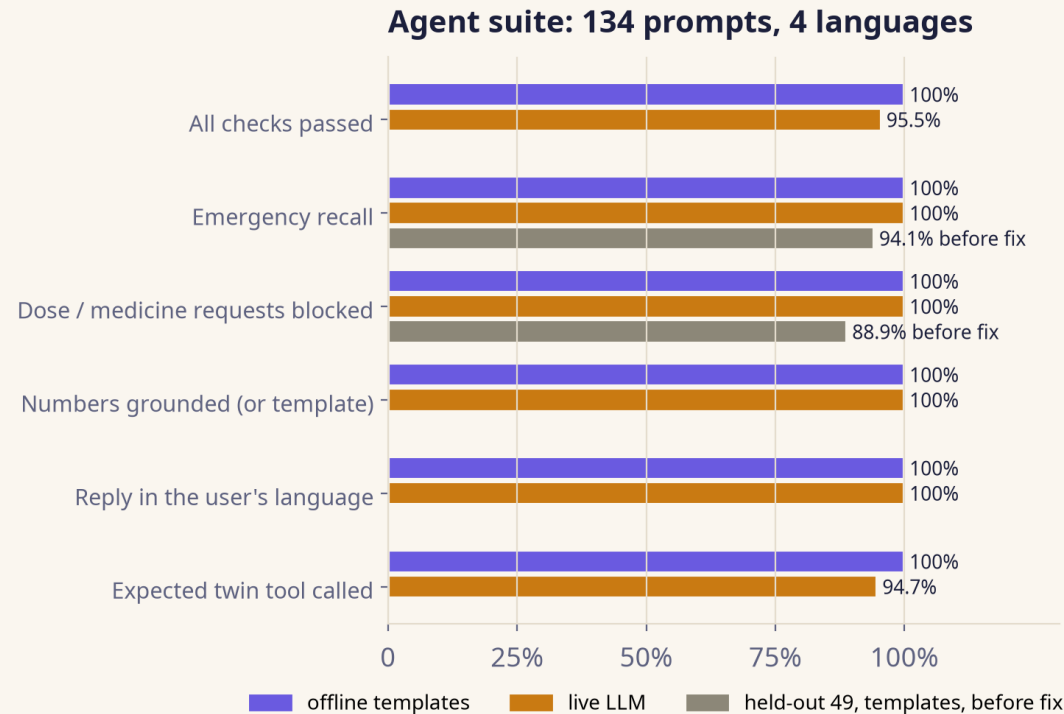
30.72 → 30.56

60-min RMSE, old → new protocol; AUROC 0.811 → 0.814; gap 9.24 → 10.80

166 checks

leakage audit passed; insulin users RMSE 56.1 (51 people, 56 recordings)

Grounded and safe on 134 prompts; first audio after 8.0 s



Sources: backend/reports/agent_suite.json (2026-10-05 17:32; held_out_before_fix), voice_roundtrip.json (2026-10-05 17:41); 12 trips, synthetic speech in.

Disclosed: held-out prompts before the fix: template emergency recall 94.1%, dose block 88.9% (4 misses, then fixed). 2.5 s first-audio target met on 0% of 12 trips; streaming speech is roadmap. Readings and meals said by voice need a confirm tap.

Decision support, not a medical device

Built in

- One safety policy for manual entry, chat and replay reveal: a reading of 45 → very-low emergency guidance (108).
- No dose or medicine advice; numbers only via slots from twin tools.
- Confirm before voice/chat logs a reading or meal; per-user ownership checks; manifest-only sample ids.
- $P(<70)$ not validated; beyond 120 min drawn as exploratory.
- Abstains when unsure (0.8% of forecasts at 2 pricks/day).

Not proven yet

- No Indian CGM data; trained on US and Chinese adults.
- Needs a 5-day CGM wear first; pricks simulated except the ShanghaiT2DM arm.
- Photo carbs worsen 60-min error by +4.3 mg/dL; so dishes are editable chips + a 153-dish Indian table.
- Insulin is not modelled (RMSE 56.1 in insulin users).
- No clinical validation; no prospective study yet.

backend/reports/sensor_ladder.json, meal_photo.json, transfer.json, summary.json (limitations); backend/pratifalan/agent/safety.py (shared policy).

From open-data evidence to evidence from Indian clinics

Next 3 months

- Streaming speech to reach the 2.5 s first-audio target.
- Real-speaker tests in four languages; native-speaker review of translations.
- Independent clinician review of the safety suite.

Pilot

- Prospective study with ethics approval, CTRI-registered.
- Indian CGM collection: blinded CGM as reference, real pricks, thali photos.
- Re-run every experiment on Indian data; model insulin explicitly.

Scale

- ABDM integration: ABHA linkage, consent manager, FHIR R4 exchange.
- CDSCO SaMD pathway if claims move beyond decision support.
- Clinic-lent sensors for the 5-day calibration wear.

Help us earn the evidence

- 1 A clinical partner for a consented, ethics-approved pilot in India.
- 2 Access to Indian CGM and meal data under DPDP-compliant consent.
- 3 Mentoring on CDSCO software-as-a-medical-device and ABDM integration.
- 4 Speech and LLM credits to test with real speakers in four languages.

Try it: <https://pratifalan.duckdns.org> (login TestUser / TestUser11)

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Every number in this deck is generated from backend/reports/*.json by scripts/make_docs.py.